Sections: B,C & D

Linear Algebra (MT-1004)

Homework#33

Instructor: Muhammad Ali

Question 1

Consider the set of functions $S = \{e^x, e^{2x}, e^{3x}\}$ defined on the interval $I = [0, 1]$. Use the **Gram-Schmidt process** to orthogonalize this set with respect to the inner product:

$$\langle f, g \rangle = \int_0^1 f(x)g(x)\,dx.$$